

Zack Dulberg

PHYSICIAN · NEUROSCIENTIST

✉ zdulberg@alumni.princeton.edu | [in zack-dulberg](https://www.linkedin.com/in/zack-dulberg)

Education

Princeton University

PHD COMPUTATIONAL NEUROSCIENCE

- Advisor: Jonathan Cohen

Princeton, NJ

2019 - 2024

University of Ottawa

RESIDENCY IN FAMILY MEDICINE

Ottawa, ON

2017 - 2019

University of Toronto

DOCTOR OF MEDICINE

Toronto, ON

2013 - 2017

McGill University

B.Sc., JOINT MAJOR IN PHYSICS AND PHYSIOLOGY

- Cumulative GPA: 3.97/4.0. Awards: Dr. Kathleen Terroux Prize in Physiology (2013, \$400), Loretta Del Bosco Scholarships in Science (2011, \$2500), Michael Wing Kin Chan & Lana Lai Yan Chan Scholarship (2010, \$2500)

Montreal, QC

2009 - 2013

Publications

Dulberg Z, Dubey R, Cohen JD (2025). Adapting to loss: A normative account of grief. *Psychological Review*.

Dulberg Z, Cohen JD (2025). On the duality of pain and pleasure processing: Why two dimensions of valence may be better than one. *bioRxiv*.

Dulberg Z, Dubey R, Berwian IM, Cohen JD (2023). Having multiple selves helps learning agents explore and adapt in complex changing worlds. *Proceedings of the National Academy of Sciences*, 120(28), e2221180120.

Webb TW, Frankland S, Altabaa A, Segert S, Krishnamurthy K, Campbell D, Russin J, Giallanza T, **Dulberg Z**, O'Reilly R, Lafferty J, Cohen JD (2023). The Relational Bottleneck as an Inductive Bias for Efficient Abstraction. *arXiv:2309.06629*.

Dulberg Z, Webb TW, Cohen JD (2021). Modelling the Development of Counting with Memory Augmented Neural Networks. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43.

Webb TW, **Dulberg Z**, Frankland SM, Petrov AA, O'Reilly RC, Cohen JD (2020). Learning Representations that Support Extrapolation. *International Conference on Machine Learning (ICML)*.

Levin D, **Dulberg Z**, Chan A, Hare G, Mazer C, Hong A (2017). A randomized-controlled trial of nabilone for the prevention of acute postoperative nausea and vomiting in elective surgery. *Canadian Journal of Anesthesia* 64(4), 385-395.

Dulberg Z, Levin D, Hong A (2015). An improvised pressurization system for arterial lines. *Canadian Journal of Anesthesia*, 62(4), 431-432.

Dulberg Z, Levin D, Hong A (2015). In reply: Another pressurization system for invasive monitoring. *Canadian Journal of Anesthesia*, 62(5), 547-547.

IN PREP

Dulberg Z, Cohen JD (2024). Using reinforcement learning to explain psychodynamics. *In prep*.

Presentations

* *presenting author*

INVITED TALKS

Summer 2025. *Using reinforcement learning to explain psychodynamics*. Invited talk: The Department of Psychology, Bar-Ilan University.

Fall 2024. *Adapting to loss: A normative account of grief*. Invited talk: The Neurobiology of Grief International Network (NOGIN) annual conference, virtual.

Winter 2022. *Modelling the development of counting*. Invited presentation: Serre lab meeting, Brown University, virtual.

Summer 2015. *Case report on rTMS induced mania*, Invited talk: Medical Imagination conference. Toronto, ON.

CONTRIBUTED PRESENTATIONS

Dulberg Z*, Dubey R, Cohen JD (2024). *Adapting to loss: A normative account of grief*. Poster highlight talk & poster: Computational psychiatry conference, Minneapolis, MN.

Dulberg Z*, Dubey R, Cohen JD (2024). *Adapting to loss: A normative account of grief*. Presentation at Princeton Neuroscience Institute annual retreat, Atlantic city, NJ

Dulberg Z*, Dubey R, Cohen JD (2024). *Adapting to loss: A normative account of grief*. Poster: Society for Digital Mental Health annual conference, virtual.

Dulberg Z*, Dubey R, Cohen JD (2022). *Modularity benefits reinforcement learning agents with competing homeostatic drives*. Poster: Conference on Reinforcement Learning and Decision Making (RLDM), Providence, RI.

Dulberg Z*, Cohen JD (2020). *Learning Canonical Transformations*. Spotlight Presentation & workshop paper: NeurIPS Workshop on BabyMind, virtual.

Dulberg Z* (2019). *Automatic sleep staging using an ambulatory electroencephalogram (EEG) device*, University of Ottawa Research, Inquiry and Opinion day. Awarded: best presentation

Dulberg Z*, Murray BJ (2017). *State space analysis of neurophysiological signals in health and illness*, Sunnybrook summer student poster competition. Awarded: 3rd prize, evaluative clinical sciences.

Professional Experience

2026 - **Concussion rehab Physician**, 360Concussion Clinic

2021 - **Mental health physician**, Maple Telemedicine

2020-2021 **Physician vaccinator**, COVID-19 Mass Vaccination Sites

2020-2021 **Cognitive Behavioral Therapy Fellow**, TeleCBT

2019 **Family physician locum**, Underserved rural communities in Ontario

2017 **Summer research assistant**, Computational research in sleep, Sunnybrook Sleep Lab, Dr. Brian Murray

2015 **Summer research assistant**, Clinical research in anesthesia, St. Mike's Hospital, Dr. Aaron Hong

2014 **Summer research assistant**, Clinical research in psychiatry, Toronto Western Hospital, Dr. Jonathan Downar

2012 **Summer research assistant**, Research in psychophysics, University of Toronto, Dr. Ken Norwich

2011 **Summer research assistant**, Research in biomedical engineering, University of Toronto, Dr. Milos Popovic

Awards, Fellowships, & Grants

2024	Travel award , Computational psychiatry conference	\$ 500
2017	Research Summer Student Award , Sunnybrook Research Institute	\$ 3,200
2012	Undergraduate student research award , Natural Sciences and Engineering Research Council of Canada	\$ 4,500
2011	Undergraduate student research award , Natural Sciences and Engineering Research Council of Canada	\$ 4,500

Teaching Experience _____

Fall 2020	NEU 202 – Intro to Cognitive Neuroscience , Teaching Assistant	<i>Princeton U</i>
Spr 2020	NEU 427 – Systems Neuroscience , Teaching Assistant	<i>Princeton U</i>
2018-2019	Clinical skills , Resident teaching 1st & 2nd year medical students	<i>U of Ottawa</i>
2013-2017	Piano performance & composition , Private teacher	<i>Toronto, ON</i>

Mentoring _____

2024	Yiming Zhao , Masters student from University of Amsterdam, summer intern	<i>Princeton University</i>
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Service & Outreach _____

2019-2024	NeurON: The Neuroscience Outreach Network , Advisory Board Member	<i>Princeton, NJ</i>
2023	Halutz fellowship , Leadership program at Ben Gurion University	<i>Beersheba, Israel</i>

Peer Review _____

Reviewer for Neural Information Processing Systems (NeurIPS) Conference, 2021
Reviewer for PLOS ONE journal, 2025

Professional Memberships _____

The College of Physicians and Surgeons of Ontario
The College of Family Physicians of Canada

Skills & Interests _____

Skills: Python; Pytorch; Reinforcement learning; Machine learning; Medicine; Psychotherapy

Interests: Piano (Royal Conservatory ARCT designation in piano performance); composition (musical director for U of T's annual musical production Daffydil); classical music (president of McGill Classical Music Club); muay thai; rock climbing.